# Cooling System – General Data

1. General

The cooling system is essentially the means of supplying ram air to the heating system, and to the cabin gaspers installed along both sides of the cabin at each seat position. Additionally, the cooling system supplies air to the gaspers in the cockpit.

2. Description

Refer to Figure 1 for details.

Components of the cooling system are: ram air scoop, main duct, main duct fan, fan duct, ram air valve, ducts and gaspers.

A. Ram Air Scoop and Main Duct

The ram air scoop is a protruding intake on the left-hand side of the fuselage nose which provides the total supply of ram air. The polycarbonate main duct connects the ram air scoop to the main duct fan.

Note For aircraft incorporating Mod 6/2387, the air scoop has rubber washers installed. Torque the screws to compress the rubber washers (which are underneath the metal washers) to approximately one-half of their original thickness. For air scoop without rubber washers (Pre Mod 6/2387) torque the #10 screws to 10 – 16 in–lbs.

B. Main Duct Fan

An electrically operated fan is installed between the main duct and fan duct, below the flight compartment floor. The fan electrical circuit (refer to 21-60-00, Environmental Control System (Mod 6/2014) – General Data) is protected by a 20–amp VENT FAN circuit breaker on the overhead circuit breaker panel, and the fan is controlled by a switch mounted on the flight compartment pedestal. Access to the fan is through a side panel, below the flight compartment left door.

C. Fan Duct

The Y–shaped fan duct connects to the main duct fan, heating system silencer inlet, and the cabin air duct. A ram air valve is installed in the heating system silencer branch of the duct.

D. Ram Air Valve

The butterfly type ram air valve, located in the fan duct, is manually operated by a control cable in the flight compartment, to regulate ram air flow to suit heating system requirements.

E. Cooling System Ducts

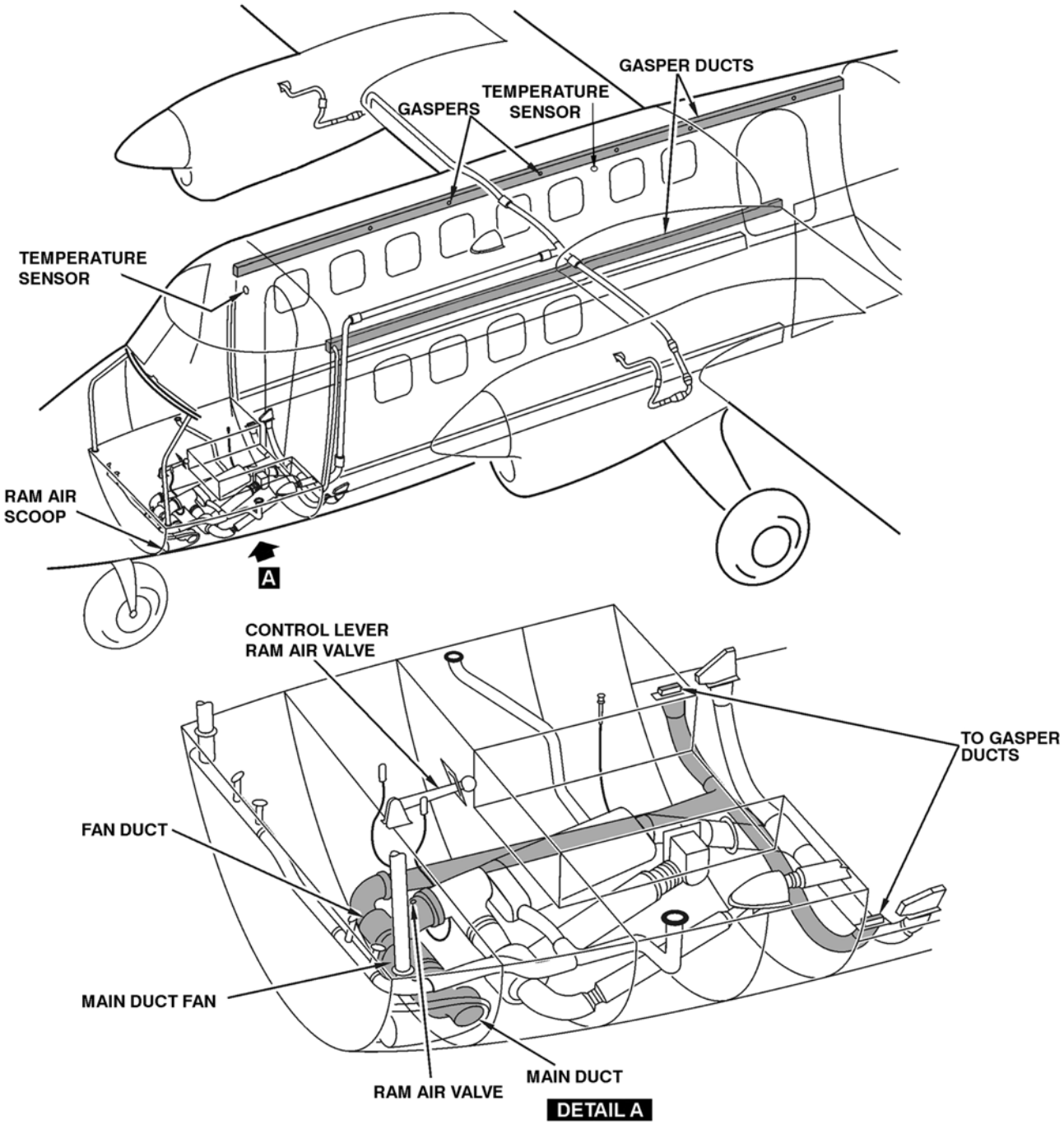
The cooling system ducting connects to the fan duct and is routed under the flight compartment floor. The cooling system ducting provides cooling for the cockpit gaspers, to the avionics bay forward of the cockpit instrument panel, and up both sides of the cabin behind the flight compartment/cabin bulkhead to join box section ducts running along both sides of the cabin above the cabin windows. The box section ducts house the cabin gaspers.

F. Cooling System Outlets

The cooling system outlets, or gaspers, are installed in the cockpit instrument panel as well as each seat position on the box section ducts.

3. Operation

In flight, ram air is ducted from the scoop on the left-hand side of the fuselage nose, through the fan, into the fan duct. The fan duct splits to supply ram air to the heating system and the cooling system. When the aircraft is stationary, the main duct fan may be switched on to draw in outside air to supply cooling air.




Cooling System Installation
Figure 1